

PART - 1

AI CONCEPTS

1. What is Artificial Intelligence?

A. Artificial Intelligence (AI) is a technology which helps computers and machines perform tasks that normally require human intelligence. It allows machines to learn from data, solve problems, make decisions, and understand information.

AI makes computers smarter so that they can do things like answer questions, recognize and generate images based on our requirement, translate languages, and give recommendations. Nowadays, AI is used in many applications such as ChatGPT, Google Assistant, YouTube recommendations, and self-driving cars.

For example, when we ask ChatGPT a question and receive an answer, AI helps the system to understand our request and respond accordingly.

AI makes many tasks easier, faster, and more efficient.

2. Difference between AI, Machine Learning, Generative AI, and Agentic AI.

A. AI:

Artificial Intelligence is a technology which enables computers and machines to perform tasks that normally require human intelligence, such as thinking, learning, and decision-making.

Example: Google Assistant and Siri.

Machine Learning:

Machine Learning is a part of AI that allows computers to learn from data and improve their performance without being explicitly programmed.

Example: YouTube video recommendations.

Generative AI:

Generative AI is a type of AI that can create new content such as text, images, videos, and code.

Example: ChatGPT generating answers to questions.

Agentic AI:

Agentic AI is an advanced type of AI that can make decisions and perform tasks independently to achieve a goal.

Example: An AI assistant that schedules meetings and sends emails automatically.

3. Explain any three open-source AI models available on Hugging Face.**A. Gemma:**

Gemma is an open-source AI model developed by Google. It is designed to be lightweight and efficient while providing strong performance in text generation and language understanding tasks.

Example: Answering questions and generating text.

Phi-4:

Phi-4 is an open-source language model developed by Microsoft. It is known for its strong reasoning abilities and efficient performance despite being smaller than many large AI models.

Example: Solving problems, answering questions, and assisting with coding tasks.

BLOOM:

BLOOM is an open-source multilingual language model created by a global team of researchers. It can understand and generate text in many different languages, making it useful for translation and content generation.

Example: Generating text in multiple languages.

4. What is the difference between traditional programming and AI-assisted development?

- A. In traditional programming, the programmer has to write all the code and instructions manually. The computer follows the rules given by the programmer and performs tasks accordingly. If there is an error, the programmer has to find and fix it on their own.

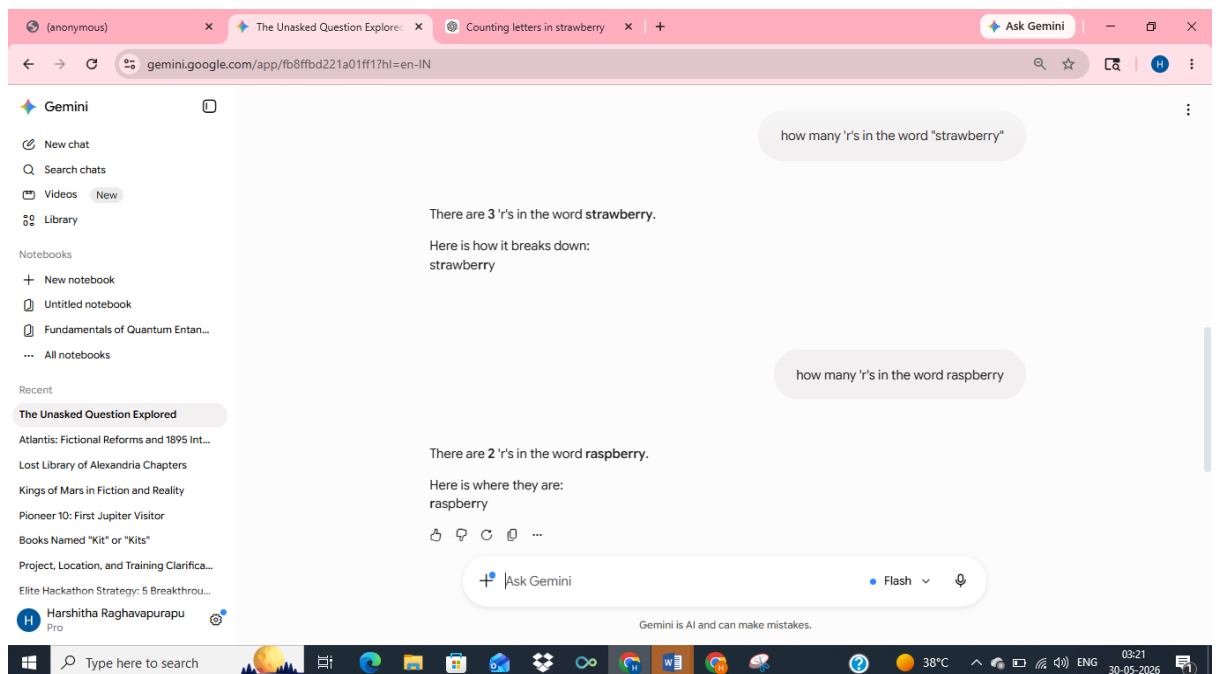
AI-assisted development uses AI tools to help developers write code, suggest solutions, detect errors, and improve productivity. Tools like ChatGPT and GitHub Copilot can assist developers by generating code and providing recommendations. This helps save time and makes the development process faster and easier.

5. Explain AI hallucinations with one example.

A. AI hallucination is a situation where an AI system provides incorrect or false information while presenting it as if it were true. This happens because AI generates responses based on patterns learned from large amounts of data rather than actually understanding facts like humans do.

There are several reasons why AI hallucinations occur. They can happen when the AI does not have enough information about a topic, when the user's question is unclear, or when there are gaps in the data used to train the AI. Sometimes the AI tries to give an answer even when it is not completely sure, which can lead to incorrect information.

Example:



AI-generated response from ChatGPT explaining Artificial Intelligence:

